

Wealth RM Copilot - Runbook, Setup Guide & Architecture

Project Overview

The Wealth RM Copilot is an enterprise-style AI assistant for Wealth Management and Relationship Managers (RMs). It supports portfolio analysis, investment research, product recommendations, suitability checks, compliance-aware responses, human-in-the-loop escalation, Retrieval Augmented Generation (RAG), LangGraph orchestration, Streamlit enterprise UI, and RAGAS evaluation metrics.

High-Level Architecture

Streamlit UI → LangGraph Agent → Intent Routing → Retrieval Layer → FAISS Vector Store → OpenAI Models → Response Generation → RAGAS Evaluation.

Folder Structure

app/, src/, evaluation/, data/, logs/, requirements_app.txt, requirements_eval.txt, .env

Setup Guide

1. Create virtual environment
2. Install dependencies
3. Configure .env
4. Run Streamlit application
5. Run RAGAS evaluation separately

LangGraph Workflow

context_guardrail → portfolio → research → suitability → human_review OR synthesize → review → END

Human-in-the-Loop (HITL)

The system detects risky investment requests and escalates them for manual review by a licensed relationship manager. Ticket IDs are generated and audit logs are maintained.

Retrieval Architecture

FAISS vector database with OpenAI embeddings, metadata filtering, client-aware retrieval, cached retriever, and MMR retrieval optimization.

RAGAS Metrics

Implemented metrics include Faithfulness, Answer Relevancy, Context Precision, and Context Recall.

Current Improvement Areas

Improve retrieval quality, semantic reranking, grounding, observability, authentication, vector database scalability, streaming responses, and evaluation frameworks.

Enterprise Features Implemented

LangGraph orchestration, multi-format ingestion, FAISS retrieval, metadata filtering, intent-aware routing, HITL escalation, ticket generation, audit logging, Streamlit UI, Pydantic validation, RAGAS evaluation, and automated test runner.

Recommended Next Enhancements

Improve RAGAS scores, add LangSmith tracing, implement streaming responses, role-based access control, portfolio visualizations, and enterprise integrations.

Production Readiness Assessment

UI: Good | RAG Pipeline: Moderate | HITL: Good | Evaluation: Moderate | Security: Basic | Scalability: Moderate | Observability: Needs Improvement | Retrieval Quality: Needs Improvement

Important Commands

```
Create App Environment:
python -m venv venv

Activate Environment:
venv\Scripts\activate

Install Dependencies:
pip install -r requirements_app.txt

Run Streamlit App:
streamlit run app/streamlit_app.py

Create Evaluation Environment:
python -m venv ragas_env

Run RAGAS Evaluation:
python evaluation/ragas_eval.py

Run Functional Tests:
python evaluation/test_run.py
```

RAGAS Metrics Summary

Metric	Purpose
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Faithfulness	Grounding quality
Answer Relevancy	Query relevance
Context Precision	Retrieval quality
Context Recall	Retrieval completeness

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